OPERATION PERFORMED: Total laparoscopic hysterectomy, bilateral salpingo-oophorectomy (TLH-BSO), and cystoscopy.

INDICATION: A 52-year-old perimenopausal female with symptomatic enlarging uterine leiomyomata, menorrhagia resulting in microcytic iron-deficiency anemia, and pelvic heaviness unresponsive to medical therapies.

OPERATIVE FINDINGS: The uterus was symmetrically enlarged to approximately 14-week gestational size (estimated weight 380 grams) with multiple subserosal and intramural fibroids, the largest measuring 6 cm in the posterior fundus. Normal bilateral ovaries and fallopian tubes without endometriosis or adhesions.

PROCEDURE DESCRIPTION: General endotracheal anesthesia was induced. The patient was positioned in dorsal lithotomy in padded Allen stirrups. A RUMI uterine manipulator with a medium Koh-efficient colpotomy ring was placed transcervically. A 10 mm infraumbilical optical trocar was placed using the direct-entry technique. Pneumoperitoneum was established to 15 mmHg. Three 5 mm accessory trocars were inserted under direct laparoscopic visualization in the bilateral lower quadrants and suprapubic midline.

Using the LigaSure bipolar vessel sealing device, the bilateral infundibulopelvic (IP) ligaments containing the ovarian vessels were identified, mobilized, desiceated, and transected, carefully protecting the retroperitoneal ureters during dissection. The round ligaments were transected, and the anterior leaf of the broad ligament opened to develop the bladder flap. The bladder was dissected and reflected off the anterior cervix down to the vaginal fornix.

The posterior leaf of the broad ligament was opened to expose the uterine vessels. The bilateral uterine arteries were skeletonized and sealed at the level of the internal cervical os. Circumferential colpotomy was executed along the Koh colpotomy cup using monopolar hook electrocautery. The uterus, cervix, fallopian tubes, and ovaries were detached and delivered intact vaginally through the vaginal vault.

The vaginal cuff was closed laparoscopically using a continuous running 0 V-Loc barbed absorbable suture in two layers, incorporating the uterosacral ligaments for apical vault support. The cuff was demonstrated to be completely hemostatic. Diagnostic cystoscopy using a 70-degree lens demonstrated intact bilateral ureteral jets with intravenous sodium fluorescein and no thermal injury or sutures within the bladder.